

Core components of MLOps	
- Data collection and preprocessing - Model selection and training - Model evaluation	The process of collecting data, selecting models, training, and evaluating their performance.
- Continuous integration (CI) - Continuous delivery (CD)	Automation of integrating, testing, and deploying ML models.
- Model monitoring - Performance metrics (accuracy, precision, recall, etc) - Model retraining when necessary	Tracking and assessing the performance of deployed models.
- Versioning and audit trails - Security and compliance (e.g., GDPR, HIPAA)	Ensuring transparency, accountability, and adherence to regulations in model management.
- Collaboration tools (communication between teams) - Automation of repetitive tasks (data preprocessing, model deployment, etc)	Facilitating teamwork and automating tasks throughout the ML lifecycle.

What is MLOps?

MLOps, or machine learning operations, is a set of practices that combines machine learning, DevOps, and data engineering principles to streamline and automate the lifecycle of machine learning models. It aims to bridge the gap between experimental model building and production deployment, ensuring that models are reliably and efficiently deployed, monitored, and updated.

Key tools and technologies used for MLOps

To support MLOps practices, several tools are widely used across the ML lifecycle.

- **TensorFlow**: An open source platform that simplifies model building, training, and deployment, with extensive support for deep learning.
- **Kubeflow**: A Kubernetes-native platform that facilitates the deployment, scaling, and management of ML workflows across various environments.
- **MLflow**: An open source platform that manages the end-to-end machine learning lifecycle, helping with model versioning, experimentation, and deployment.

What is AIOps?

AIOps, or artificial intelligence for IT operations, leverages the power of AI, specifically machine learning, to automate and improve various aspects of IT operations management. It moves beyond traditional rule-based systems by using algorithms to analyse massive amounts of IT data, identify patterns, predict issues, and automate responses, ultimately leading to more efficient and reliable IT systems.

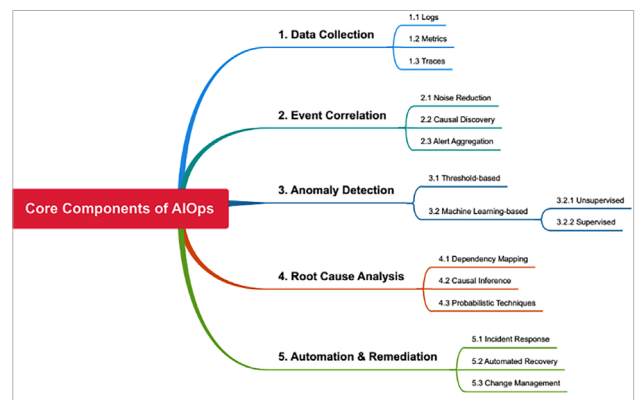


Figure 2: Core components of AIOps

Core components of AIOps	
Data collection	Gathering data from various sources (logs, metrics, and events) for analysis.
Event correlation	Identifying patterns and reducing noise to pinpoint issues more accurately.
Anomaly detection	Detecting deviations from normal behaviour to highlight potential problems early.
Root cause analysis	Using AI to find the underlying cause of incidents for faster resolution.
Automation and remediation	Automating issue resolution based on learned behaviours, reducing manual intervention.
Predictive analytics	Forecasting potential future issues by analysing historical data.